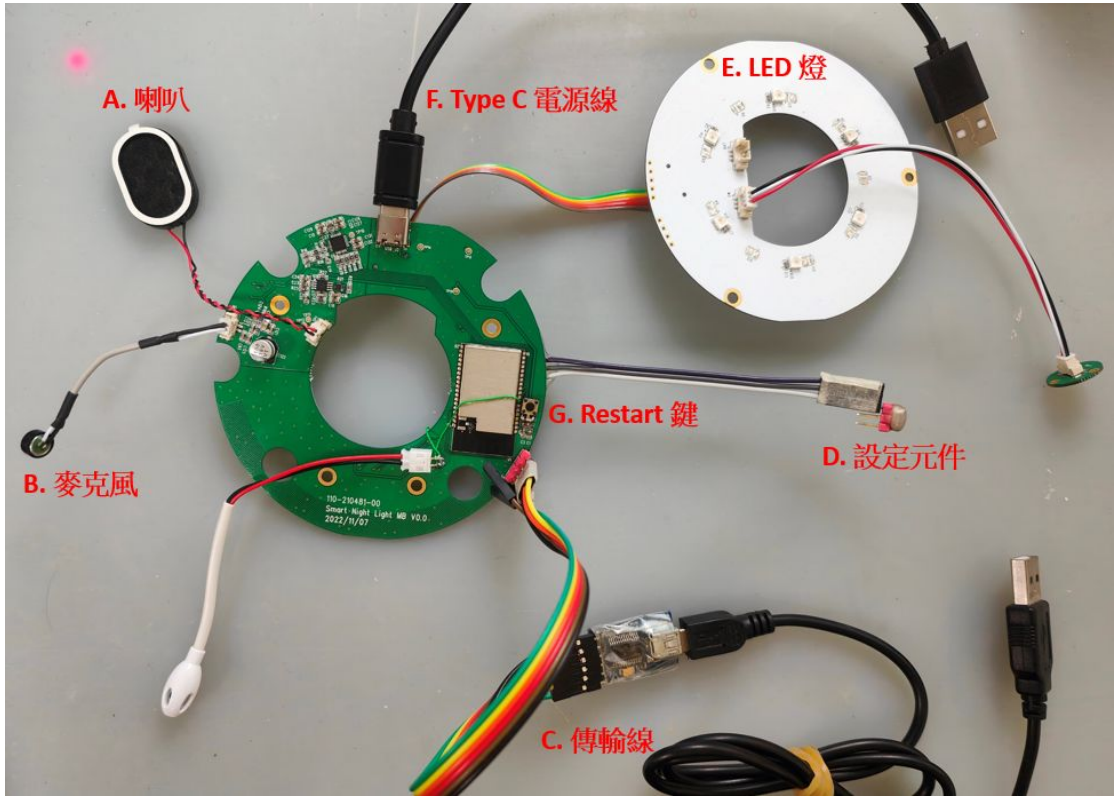
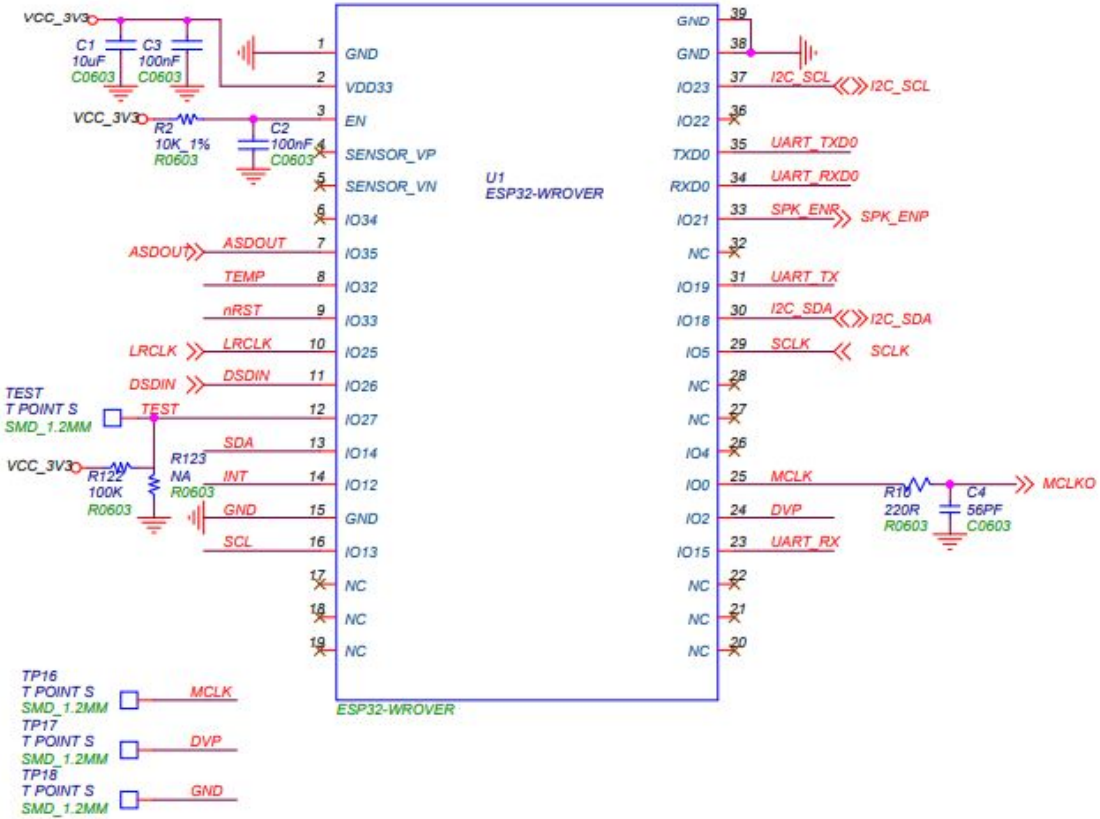
Smart Speaker Platform

平台主板



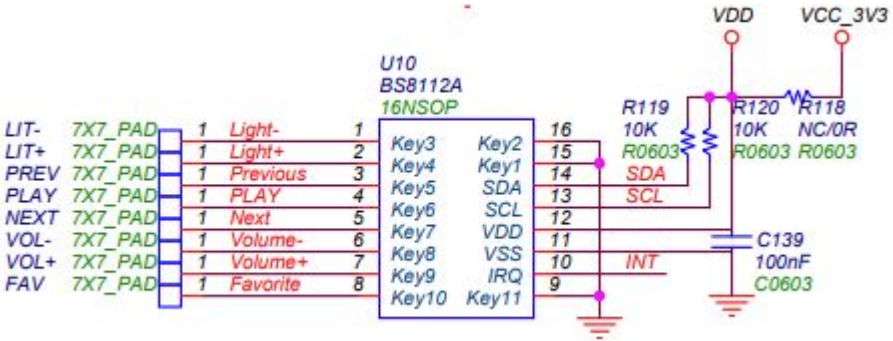
A. 喇叭:機器輸出語音
B. 麥克風:接收使用者指令
C. 傳輸線:燒錄程式碼
D. 設定元件:進入上傳模式
E. LED燈:
F. Type-C 電源線:提供電源
G. Restart鍵:重啟機器



- 核心晶片:Espressif ESP32-D0WDQ6/ESP32-D0WD (雙核 240MHz MCU)。
- 儲存配置:
  - Flash:標準 4MB SPI Flash，部分型號可達 16MB。
  - PSRAM:內建 4MB/8MB PSRAM。
- 無線技術:
  - Wi-Fi:802.11 b/g/n (最高 150 Mbps)。
  - 藍牙:Bluetooth 4.2 BR/EDR 與低功耗藍牙 (BLE)。
- 天線選擇:
  - PCB 板載天線 (如 ESP32-WROVER, ESP32-WROVER-E)。
  - IPEX 接頭外接天線 (如 ESP32-WROVER-I, ESP32-WROVER-IE)。
- 介面與外設:GPIO、UART、SPI、I2C、I2S、ADC、DAC、PWM、SD 卡接口、電容式觸控、霍爾傳感器。

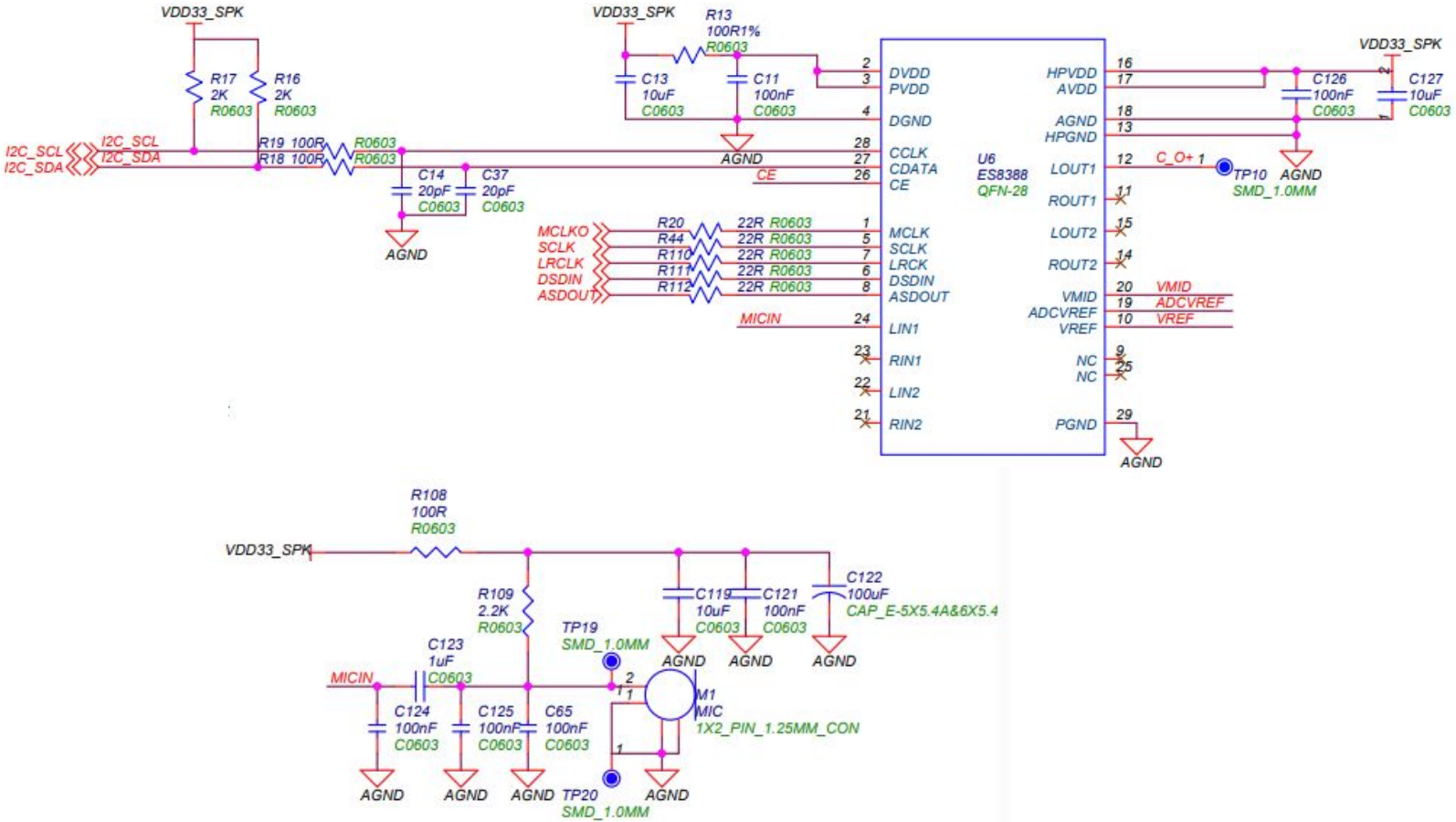
Touch IC

目前在原始碼中沒有它的驅動程式。

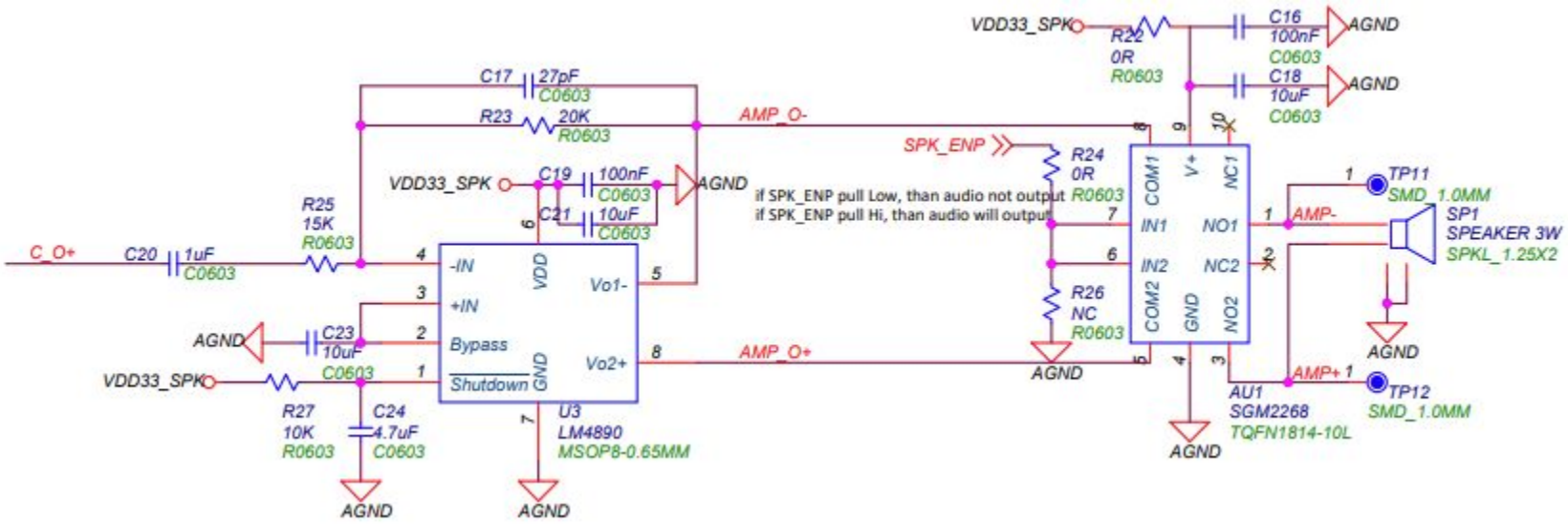


[bs8112 datasheet]

audio input

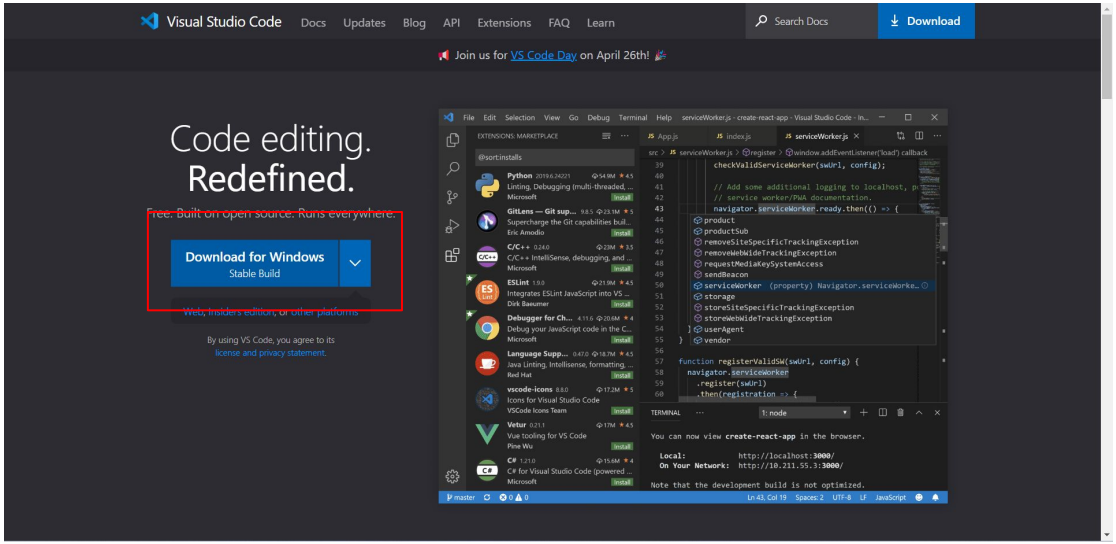

VDD33_SPK
R17 2K R0603
R16 2K R0603
I2C_SCL
I2C_SDA
R19 100R R0603
R18 100R R0603
C14 20pF C0603
C37 20pF C0603
AGND
R13 100R1% R0603
C13 10uF C0603
C11 100nF C0603
MCLKO
SCLK
LRCLK
DSDIN
ASDOUT
R20 22R R0603
R44 22R R0603
R110 22R R0603
R111 22R R0603
R112 22R R0603
MICIN
CE
U6 ES8388 QFN-28
DVDD
PVDD
DGND
CCLK
CDATA
CE
MCLK
SCLK
LRCK
DSDIN
ASDOUT
LIN1
RIN1
LIN2
RIN2
HPVDD
AVDD
AGND
HPGND
LOUT1
ROUT1
LOUT2
ROUT2
VMID
ADCVREF
VREF
NC
NC
PGND
C_O+ 1
TP10 SMD_1.0MM
C126 100nF C0603
C127 10uF C0603
VMID
ADCVREF
VREF
R108 100R R0603
R109 2.2K R0603
C123 1uF C0603
C124 100nF C0603
C125 100nF C0603
C65 100nF C0603
TP19 SMD_1.0MM
TP20 SMD_1.0MM
M1 MIC 1X2_PIN_1.25MM_CON
C119 10uF C0603
C121 100nF C0603
C122 100uF CAP_E-5X5.4A&6X5.4

audio output



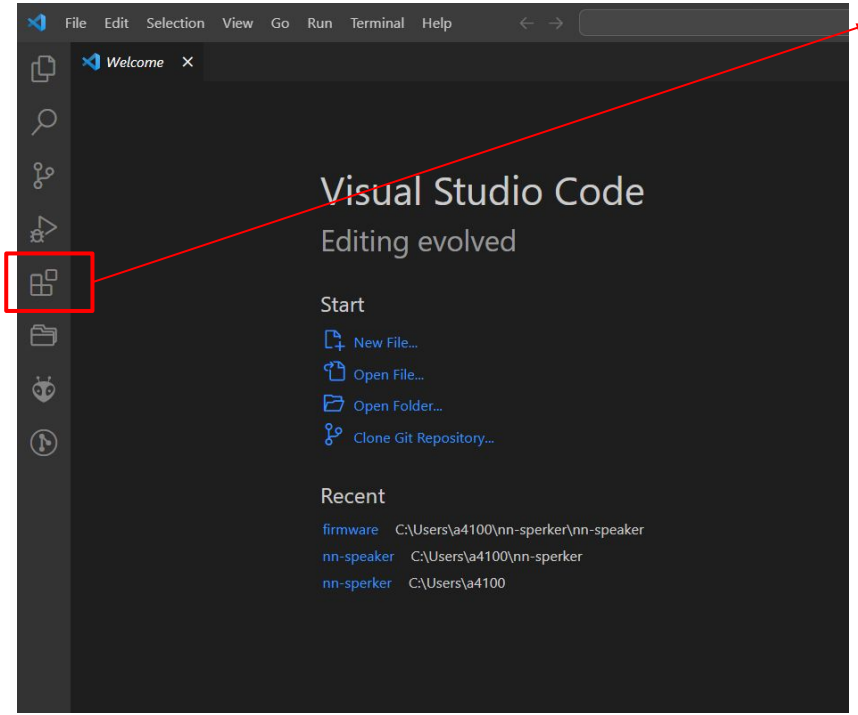
安裝作業環境-安裝 VSCode

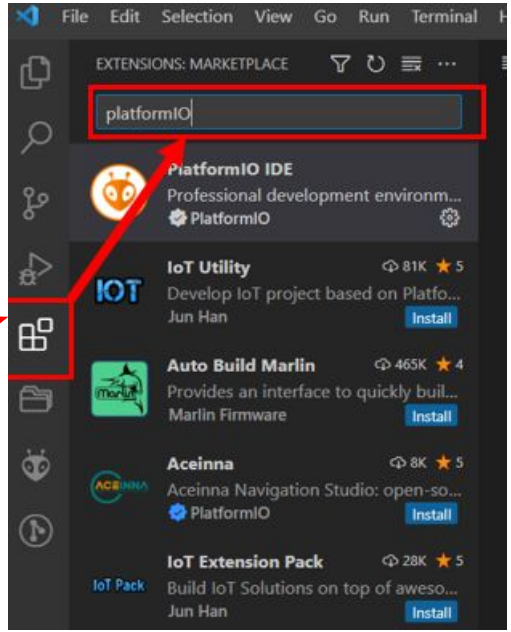
https://code.visualstudio.com/



如使用雲端平台，只要遠端 remote-desktop 就可以了。VSCode 已預裝在桌面上

安裝 platform IO



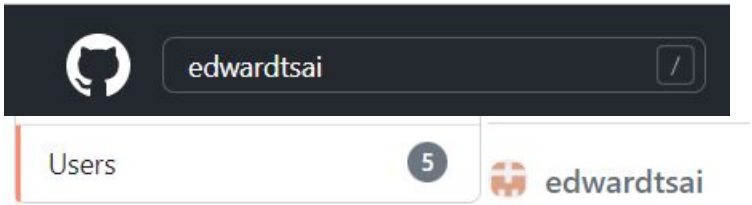


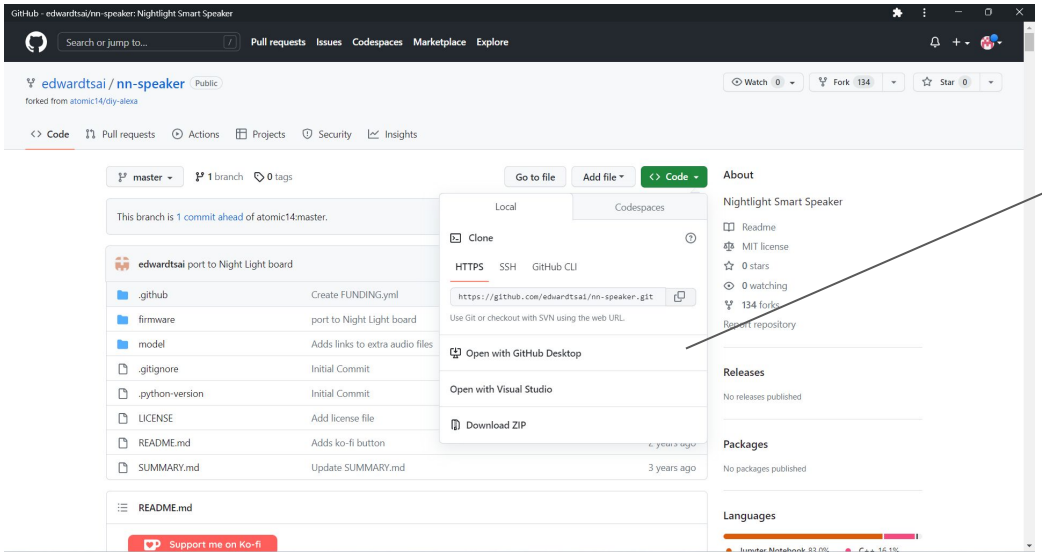
Platform IO 可以用來開發各式 MCU 的程式。

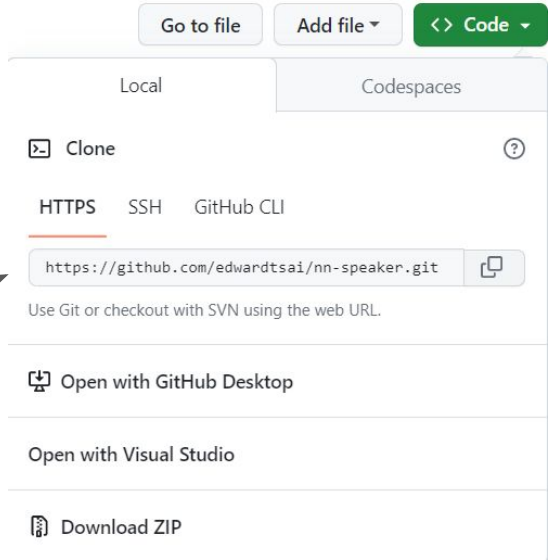
請選擇應用程式列表，然後使用 platformio 做關鍵字找出 platformIO 模組安裝。

download nn-speaker

https://github.com/wycc/nn-speaker.git

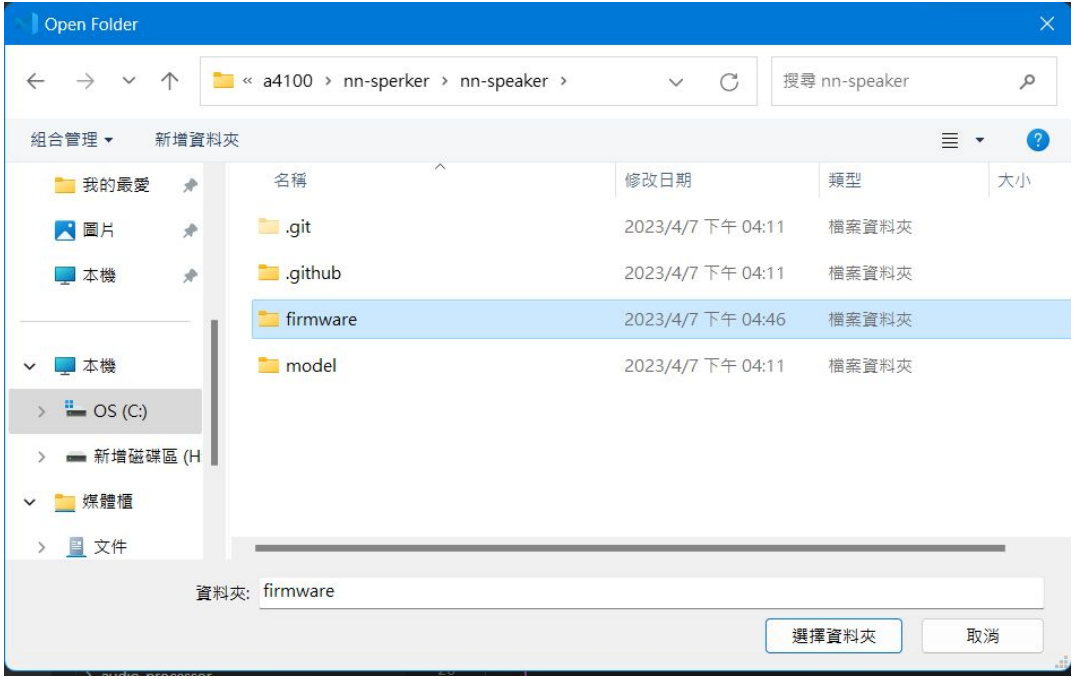




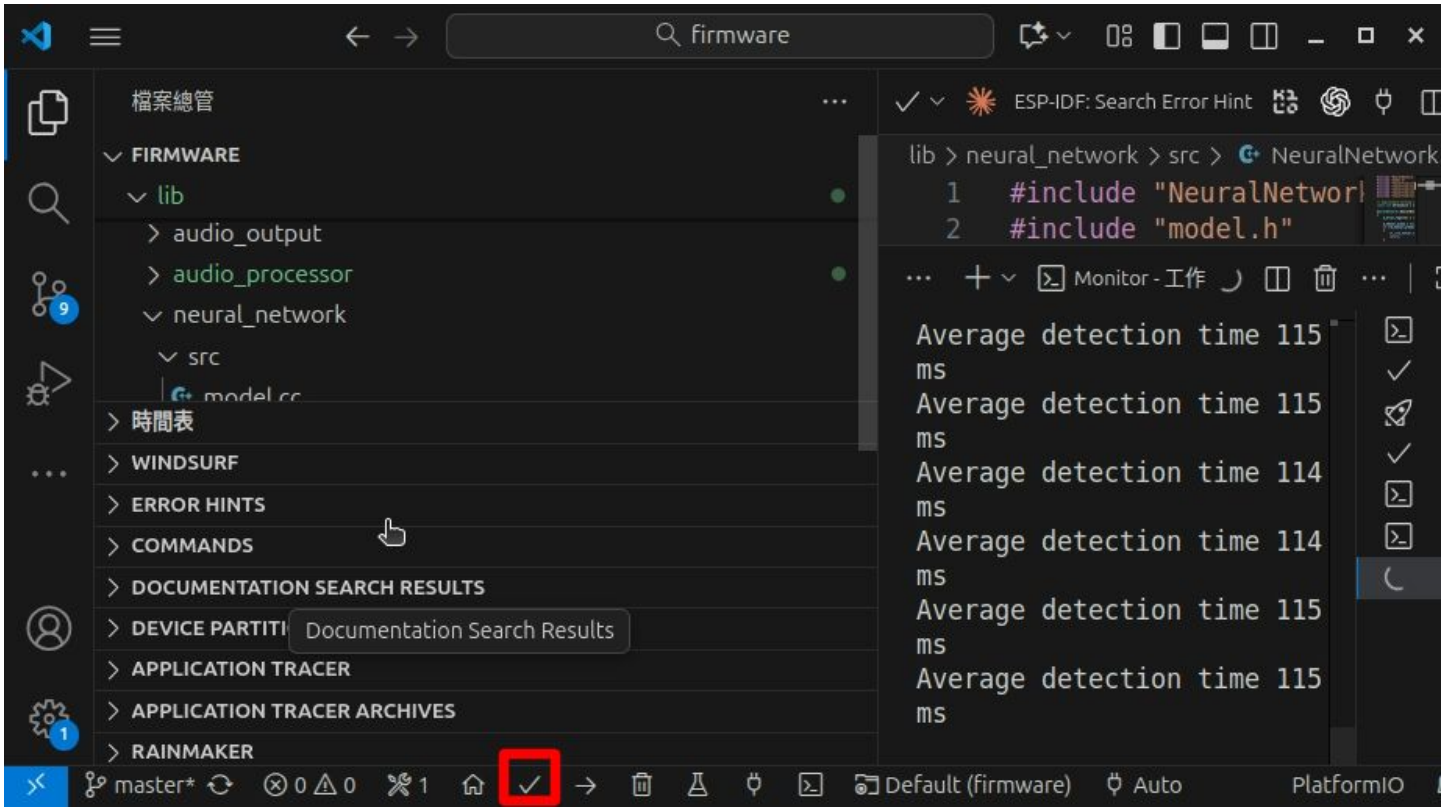


Open firmware folder

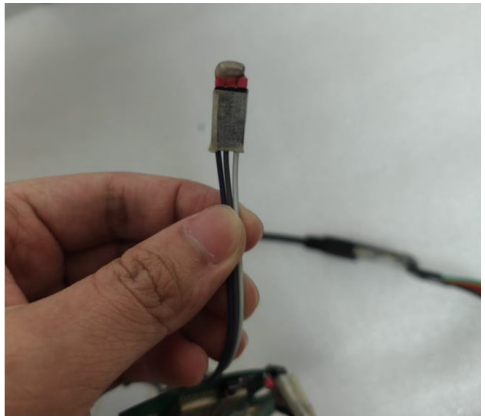
在Visual Studio Code裡開起nn-speaker 資料夾中的firmware資料夾



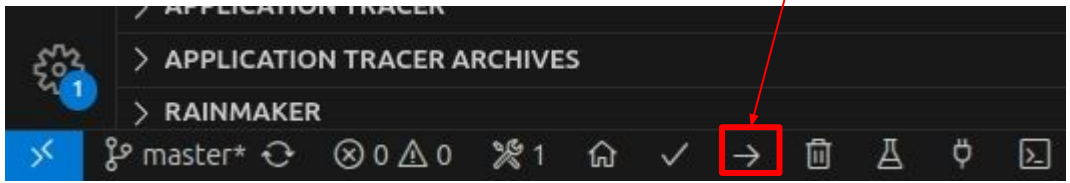
Compile the program



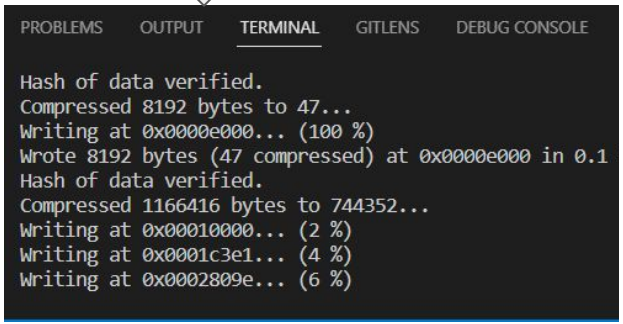
下載程式



進入下載模式前，先將機器上3pin線短路在一起，進入設定模式，再重新上電，回到Platform IO按Upload，就會開始上傳新的程式

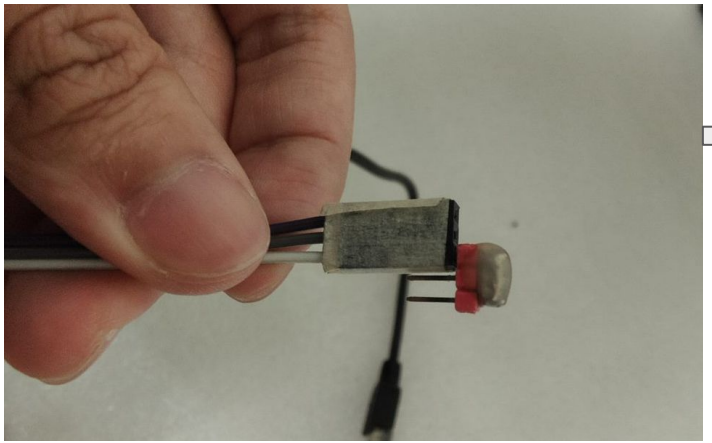


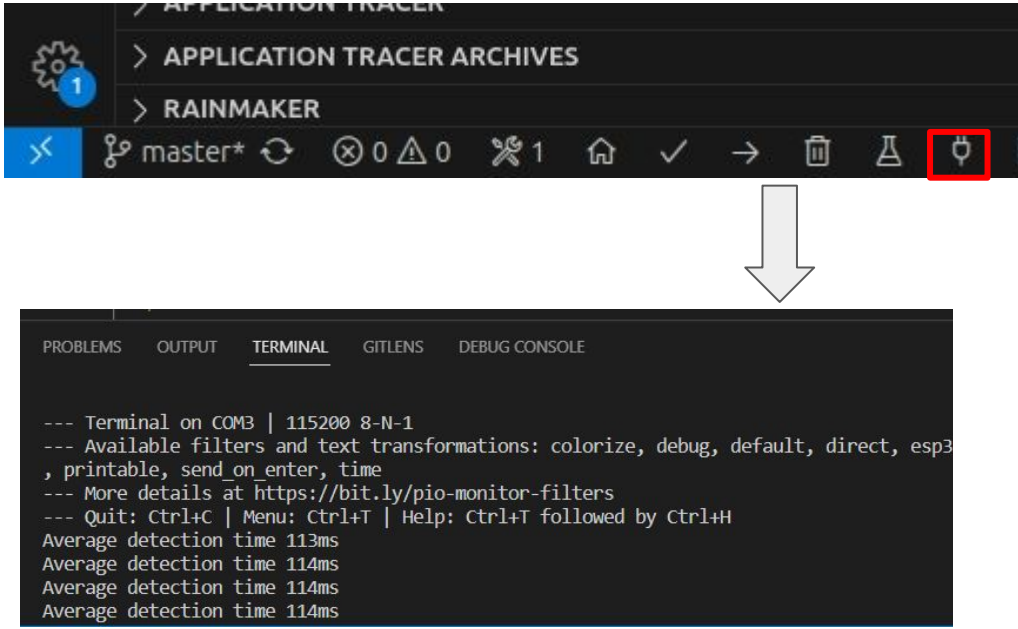
下載完後將 3pin移除



重啟系統

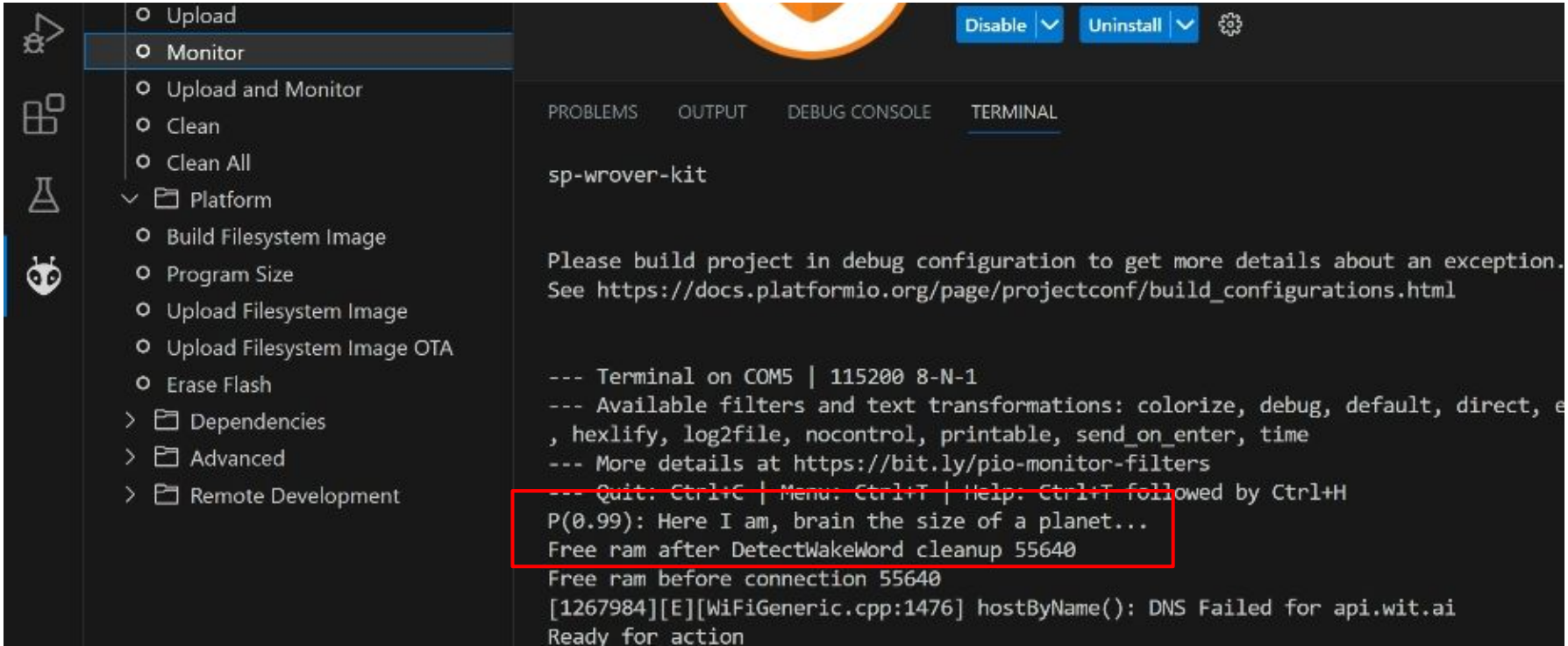
Upload 完成之後再將剛剛短路在一起的
3pin線分開, 再重新上電





測試

之後再開啟Platform IO 按下Monitor開始感測，說出Marvin, 機器就會逼一聲。



Step 1: convert all audio files into spectogram

- https://github.com/edwardtsai/nn-speaker/blob/master/model/Generate%20Training%20Data.ipynb
- Conver the audio into mel spetrogram take long time. We will do it once only.
- Therefore, we will save the files into
  - training_spectrogram.npz
  - validation_spectrogram.npz
  - test_spectrogram.npz

Step 2: Train the model

- https://github.com/edwardtsai/nn-speaker/blob/master/model/Train%20Model.ipynb
- trained.model
- fully_trained.model

```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv_layer1 (Conv2D)         (None, 99, 43, 4)         40
_________________________________________________________________
max_pooling1 (MaxPooling2D)  (None, 49, 21, 4)         0
_________________________________________________________________
conv_layer2 (Conv2D)         (None, 49, 21, 4)         148
_________________________________________________________________
max_pooling2 (MaxPooling2D)  (None, 24, 10, 4)         0
_________________________________________________________________
flatten (Flatten)            (None, 960)               0
_________________________________________________________________
dropout (Dropout)            (None, 960)               0
_________________________________________________________________
hidden_layer1 (Dense)        (None, 40)                38440
_________________________________________________________________
output (Dense)               (None, 1)                 41
=================================================================
Total params: 38,669
Trainable params: 38,669
Non-trainable params: 0
```

Step 3: convert to tflite model

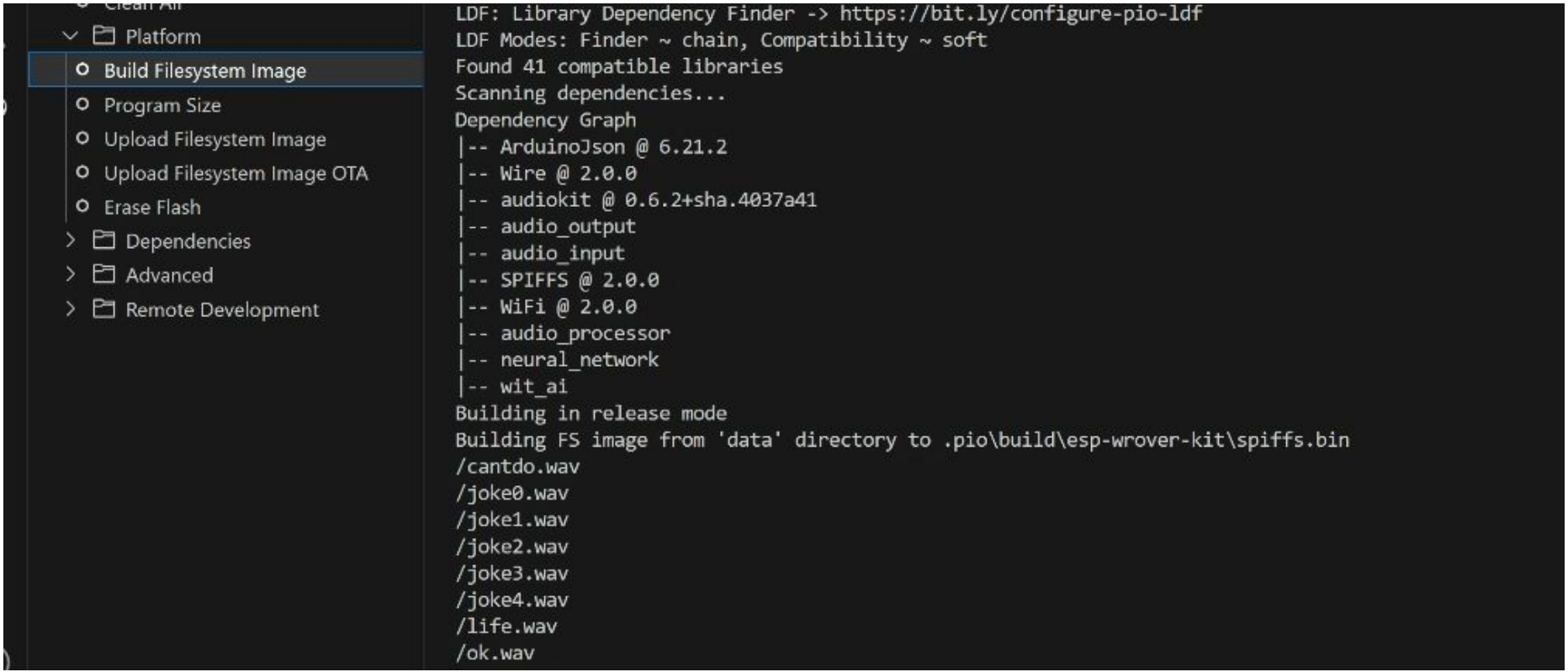
https://github.com/edwardtsai/nn-speaker/blob/master/model/Convert%20Trained%20Model%20To%20TFLite.ipynb

- Convert to tensorflow lite model
- convert the model into C source code so that we can link it into the firmware for the ESP32.

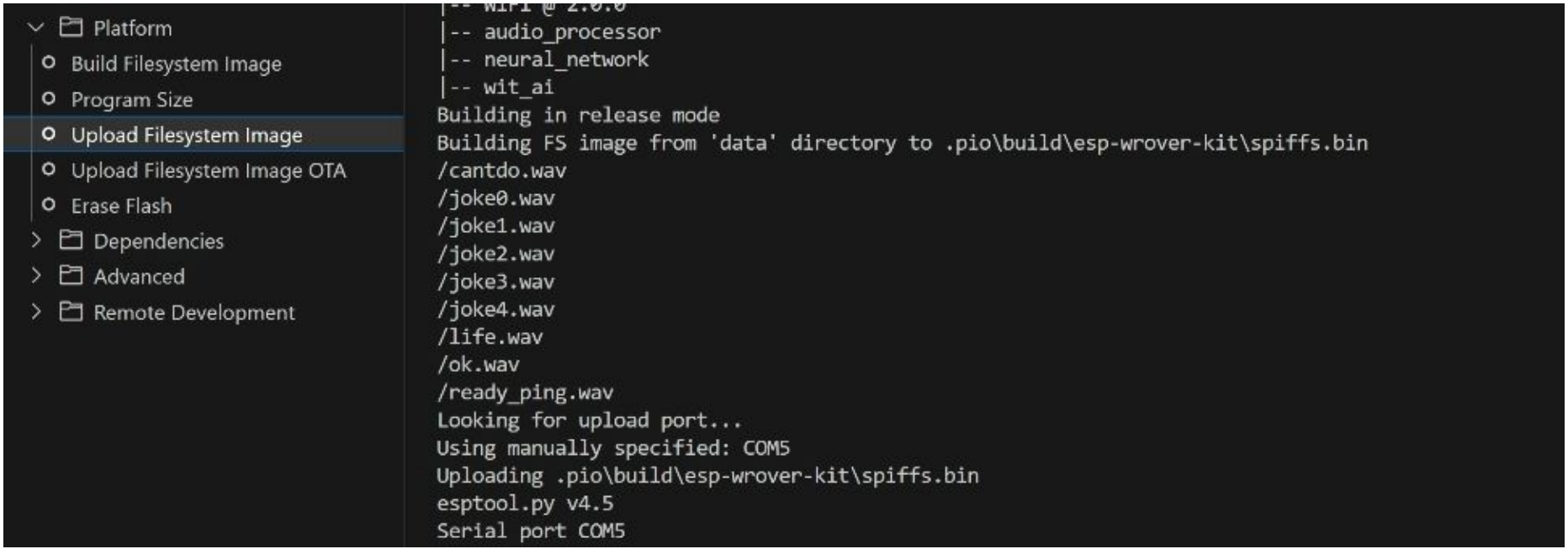
下載新的 firmware

請將之前訓練好的模型中 model_data.cc 取代原先 nn-speaker 中的同名檔案。重新編輯後下載。就可以測試新的關鑑字了。

Prepare filesystem



download filesystem



We only need to download once unless we change the files.

output sound effect

```
void RecogniseCommandState::enterState()
{
    // indicate that we are now recording audio
    m_indicator_light->setState(ON);
    m_speaker->playReady();
```

m_speaker can be used to output audio to the I2S speaker

We can play wav file

We can replace those files to whatever we need.

Play the file

```
void Speaker::playOK()
{
    m_ok->reset();
    m_i2s_output->setSampleGenerator(m_ok);
}
```

Play the wav file

The LED control

Home | Extension: C/C++ Extension Pack | IndicatorLight.cpp

IndicatorLight.cpp

```
// This task does all the heavy lifting for our application
void indicatorLedTask(void *param)
{
    IndicatorLight *indicator_light = static_cast<IndicatorLight *>(param);
    const TickType_t xMaxBlockTime = pdMS_TO_TICKS(100);
    while (true)
    {
        // wait for someone to trigger us
        uint32_t ulNotificationValue = ulTaskNotifyTake(pdTRUE, xMaxBlockTime);
        if (ulNotificationValue > 0)
        {
            switch (indicator_light->getState())
            {
            case OFF:
            {
                ledcWrite(0, 0);
                uart2_send("{8701fe}"); // off
                break;
            }
            case ON:
            {
                ledcWrite(0, 255);
                uart2_send("{8701ff}"); // on
                break;
            }
```

We can use the uart2_send to control the LED.

{8701fe} => off
{8701ff} => on

color control ???

brightness control ???

Increase model size

https://github.com/atomic14/voice-controlled-robot/blob/main/model/Train%20Model-All%20Words.ipynb

- This is an example.
- We increase the classes to be 5. Therefore, the size of dense layer is increased as well.

Layer (type)	Output Shape	Param #
conv_layer1 (Conv2D)	(None, 99, 43, 4)	40
max_pooling1 (MaxPooling2D)	(None, 49, 21, 4)	0
conv_layer2 (Conv2D)	(None, 49, 21, 4)	148
max_pooling3 (MaxPooling2D)	(None, 24, 10, 4)	0
flatten_1 (Flatten)	(None, 960)	0
dropout_2 (Dropout)	(None, 960)	0
hidden_layer1 (Dense)	(None, 80)	76880
dropout_3 (Dropout)	(None, 80)	0
output (Dense)	(None, 5)	405

Total params: 77,473
Trainable params: 77,473
Non-trainable params: 0

Please MP3 file

https://github.com/atomic14/esp32-play-mp3-demo/blob/main/src/minimp3.h

- It provide a simple MP3 decoder in a single file.
- We can use MP3 instead of wav to play longer audio file.